BIO 005 Week 9 · The Heart as a Pump

Smooth and cardiac muscle, the pacemaker, the wiring, the ECG, the cycle, and what sets cardiac output. Fill in as we go. Small square by a picture: scan it to see that picture full size on your phone.

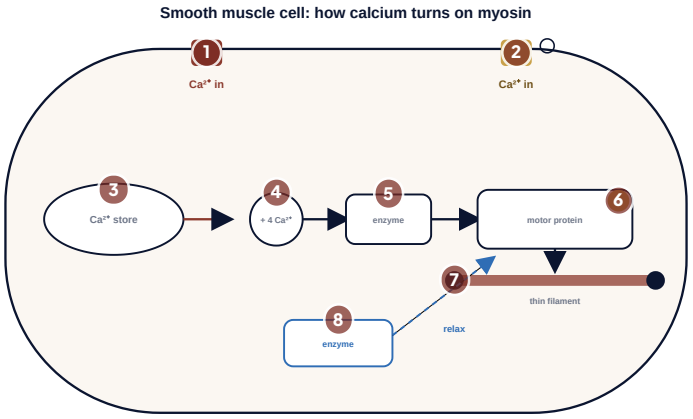
Name _____

A. Smooth muscle: how it contracts and what controls it

YOU SHOULD BE ABLE TO

Smooth muscle contraction mechanism. Trace smooth muscle contraction through calcium entry and release to calmodulin and myosin light chain kinase and explain why the mechanism is slower and more economical than in skeletal muscle.

Smooth muscle types and regulation. Compare single unit and multi unit smooth muscle and list the stimuli that alter smooth muscle tone including stretch and local chemical signals.



Smooth muscle cell: the calcium switch

1. Voltage-gated Ca^{2+} channel
2. Receptor-operated Ca^{2+} channel
3. Sarcoplasmic reticulum (smooth muscle)
4. Calmodulin
5. Myosin light chain kinase (MLCK)
6. Phosphorylated myosin light chain
7. Dense bodies
8. Myosin light chain phosphatase

WHAT YOU ARE LOOKING AT

One smooth muscle cell, cut open. The membrane is the outline, the small Ca^{2+} store (SR) sits inside, and the contractile proteins are at the lower right. There is no troponin and no sarcomere, so the switch is different: it is on myosin, and it runs through an enzyme.

Extra notes and drawings

BIO 005 Week 9 · The Heart as a Pump

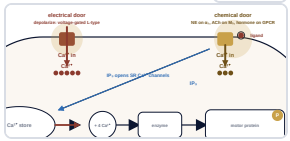
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Name _____

A. Smooth muscle: how it contracts and what controls it (continued)

HOW IT HAPPENS, IN ORDER (NUMBERS MATCH THE GOLD MARKERS ON THE SLIDE)

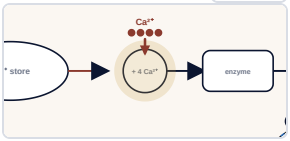
STEP 1



Ca^{2+} rises, by one of two doors

Electrical door: the cell depolarizes (stretch, a pacemaker cell, or a neurotransmitter opening a cation channel) and voltage-gated L-type Ca^{2+} channels open; Ca^{2+} enters from the ECF. Chemical door: a neurotransmitter or hormone binds a G protein-coupled receptor (norepinephrine on α_1, acetylcholine on M_3, angiotensin II, oxytocin). The receptor activates phospholipase C, IP_3 is made, IP_3 opens Ca^{2+} channels on the SR, and ligand-gated (receptor-operated) channels in the membrane let more Ca^{2+} in. No action potential is needed

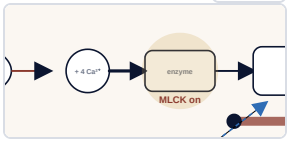
STEP 2



Ca^{2+} binds calmodulin

Four Ca^{2+} per calmodulin, in the cytoplasm. This is the smooth muscle version of Ca^{2+} meeting troponin; there is no troponin here

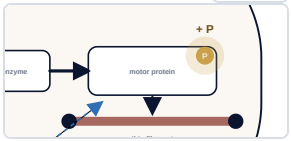
STEP 3



Ca^{2+}–calmodulin activates myosin light chain kinase

MLCK is an enzyme that was sitting idle; Ca^{2+}–calmodulin binds it and turns it on. Because this is an enzyme step, it takes time, which is one reason smooth muscle is slow to start

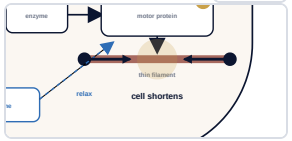
STEP 4



MLCK phosphorylates the myosin light chain

MLCK moves a phosphate from ATP onto the regulatory light chain of myosin. That phosphate switches on myosin ATPase; unphosphorylated myosin cannot bind actin

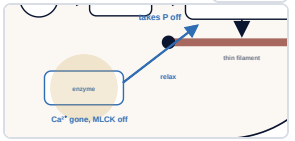
STEP 5



Crossbridges cycle and the cell shortens

Phosphorylated myosin binds actin and pulls it toward the dense bodies, so the whole cell draws in like a drawstring. Cycling is 10 to 100 times slower than skeletal muscle

STEP 6



Ca^{2+} falls and myosin phosphatase turns it off

Ca^{2+} is pumped out of the cell and back into the SR; calmodulin lets go of MLCK. Myosin light chain phosphatase removes the phosphate, so cycling stops. If a crossbridge is still attached when the phosphate comes off it can stay attached (latch state), holding tension for almost no ATP. Signals that inhibit the phosphatase (through Rho kinase) raise tension without raising Ca^{2+}

THE EFFECT: WHAT CHANGES BECAUSE OF THIS

Slow, graded contraction that can be held for hours as tone. Two doors let Ca^{2+} in: the voltage door (stretch, pacemaker slow waves, high K^+, a depolarizing transmitter) and the receptor door (a ligand on a G protein–coupled receptor, IP_3, no action potential needed); single-unit muscle leans on the first, multi-unit on the second, and most tissues have both. Tone changes whenever Ca^{2+} entry changes through either door or whenever the phosphatase is turned up or down. That is how norepinephrine constricts an arteriole (α_1, Ca^{2+} up), epinephrine relaxes a bronchiole (β_2, cAMP inhibits MLCK), nitric oxide relaxes a vessel (cGMP lowers Ca^{2+} and activates the phosphatase), and calcium channel blockers lower blood pressure.

Extra notes and drawings

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A. Smooth muscle: how it contracts and what controls it (continued)

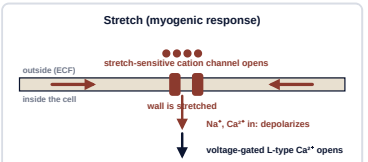
WHY SMOOTH MUSCLE IS SLOWER AND CHEAPER TO RUN

Regulation is on	**myosin (thick filament), through an enzyme step, not a troponin switch on actin**
Myosin ATPase is	**slow: 10 to 100 times slower crossbridge cycling than skeletal muscle**
The latch state	**crossbridges stay attached after dephosphorylation, holding tension with almost no ATP**
Ca^{2+} mostly comes from	**the ECF, so contraction can be graded by how much Ca^{2+} enters**

WHAT CHANGES SMOOTH MUSCLE TONE?

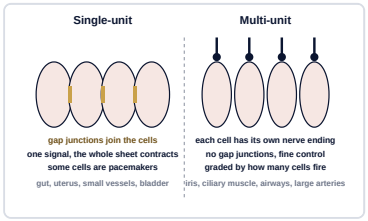
Smooth muscle tone

Stretch



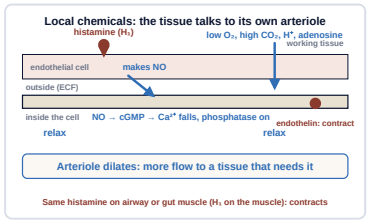
Myogenic response: stretch opens channels, the cell depolarizes and contracts

Single-unit vs multi-unit



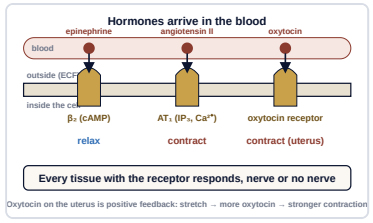
Single-unit: cells joined by gap junctions contract together, some are pacemakers (gut, uterus, small vessels). Multi-unit: each cell has its own nerve supply, fine control (iris, airways, large arteries)

Local chemicals

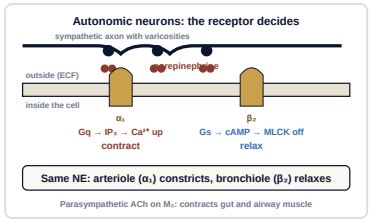


In arterioles: low O_2, high CO_2, H^+, adenosine and nitric oxide relax it; histamine relaxes it too (H_1 on the endothelium releases nitric oxide, H_2 on [illegible] which is the hypotension of [illegible] endothelin tightens it. The [illegible]ine contracts airway and gut [illegible]cle (H_1 on the muscle): [illegible]sm in the same patient

Hormones arrive in the blood



Autonomic neurons



BIO 005 Week 9 · The Heart as a Pump

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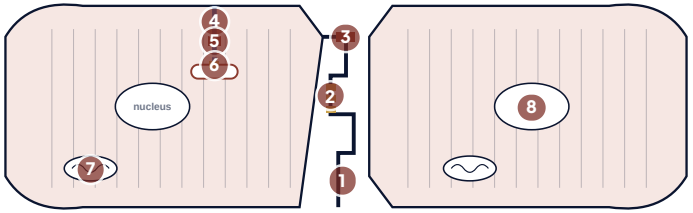
B. Cardiac muscle, and the three muscle types side by side

YOU SHOULD BE ABLE TO

Cardiac muscle properties. Explain how intercalated discs and gap junctions allow the myocardium to act as a functional syncytium and state why cardiac muscle cannot be tetanized.

Comparison of the three muscle types. Build a comparison of skeletal, cardiac, and smooth muscle by calcium source, regulatory protein, control, and speed of contraction.

Two cardiac muscle cells joined at an intercalated disc



1. Intercalated disc
2. Gap junction
3. Desmosome
4. T-tubule (cardiac)
5. L-type Ca^{2+} channel
6. Ryanodine receptor (cardiac SR)
7. Mitochondrion (cardiac)
8. Branching, single nucleus

WHAT YOU ARE LOOKING AT

Two cardiac muscle cells end to end. The stepped dark line between them is the intercalated disc, with its gap junction channels and desmosome anchors. Inside the left cell, a T-tubule drops in from the membrane and meets the sarcoplasmic reticulum.

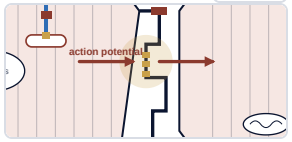
Two cardiac cells and the intercalated disc

Extra notes and drawings

B. Cardiac muscle, and the three muscle types side by side (continued)

HOW IT HAPPENS, IN ORDER (NUMBERS MATCH THE GOLD MARKERS ON THE SLIDE)

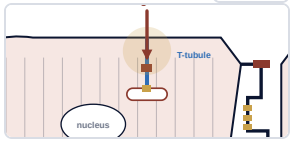
STEP 1



An action potential arrives from the neighbor

Positive current flows through the gap junction channels (connexons) in the intercalated disc and depolarizes this cell to threshold. No nerve and no neurotransmitter are involved between cardiac cells

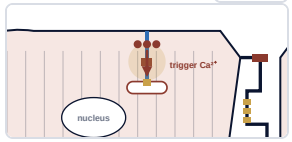
STEP 2



The signal runs down the T-tubule

Deep into the cell, to the L-type Ca^{2+} channels

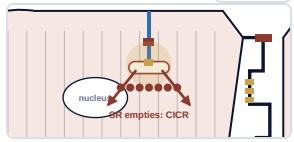
STEP 3



Trigger Ca^{2+} enters

The L-type channel opens during the plateau and a little Ca^{2+} comes in from the ECF

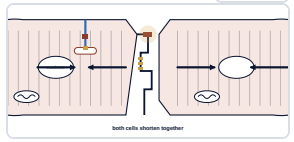
STEP 4



The SR releases the rest: calcium-induced calcium release

Trigger Ca^{2+} opens the ryanodine receptors and the store empties into the cytoplasm

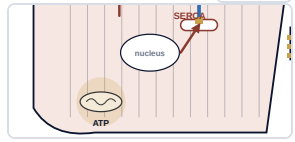
STEP 5



Both cells contract together, held by the desmosomes

Ca^{2+} binds troponin C, tropomyosin moves off the myosin-binding sites on actin, crossbridges cycle; force is graded by how much Ca^{2+} was released. The desmosomes keep the cells from tearing apart under that pull

STEP 6



Ca^{2+} is pumped away and the cells relax

SERCA refills the SR and the Na^{+}-Ca^{2+} exchanger pushes Ca^{2+} out; mitochondria pay for all of it

THE EFFECT: WHAT CHANGES BECAUSE OF THIS

The whole chamber contracts as one unit, with force graded by how much Ca^{2+} was released. Lose the gap junctions and the heart cannot act as one; lose the oxygen supply and the mitochondria stop and the cells die, which is a heart attack.

COMPARE THE THREE MUSCLE TYPES

	SKELETAL	CARDIAC	SMOOTH
Ca^{2+} source	sarcoplasmic reticulum only	ECF through L-type channels, then the SR (calcium-induced calcium release)	mostly ECF, some SR
Regulatory protein	troponin, on the thin filament	troponin, on the thin filament	calmodulin and MLCK, on the thick filament
Control	somatic motor neuron only; no nerve, no contraction	myogenic (SA node pacemaker); rate and force adjusted by autonomic nerves and hormones	autonomic nerves, hormones, local chemicals, stretch; some pacemakers
Speed and tetanus	fastest; can summate and tetanize	intermediate; cannot tetanize (long refractory period)	slowest; sustained tone at low energy cost
Cell shape	long cylinder, many nuclei at the edge, striated	short, branched, one central nucleus, striated, intercalated discs	spindle, one nucleus, no striations

C. The adrenal medulla: a sympathetic ganglion that secretes into blood

YOU SHOULD BE ABLE TO

Adrenal medulla. Explain why the adrenal medulla is a modified sympathetic ganglion and compare the duration of its circulating catecholamine effect with direct sympathetic innervation.

FROM THE SPINAL CORD TO EVERY TISSUE

STEP 1
Sympathetic preganglionic neuron fires
Leaves the thoracic cord and releases acetylcholine onto nicotinic receptors, like any sympathetic preganglionic neuron

STEP 2
Chromaffin cells respond
These are modified postganglionic neurons with no axons; that is why the medulla counts as a ganglion

STEP 3
Catecholamines enter the blood
About 80% epinephrine, 20% norepinephrine, released as hormones

STEP 4
Hormone reaches every tissue with adrenergic receptors
Including cells with no sympathetic nerve at all, such as skeletal muscle arterioles with β_2 receptors and liver cells

STEP 5
Effect lasts minutes, not milliseconds
Cleared slowly by the liver; a sympathetic nerve signal ends in milliseconds by reuptake at the varicosity

NERVE VERSUS HORMONE

Direct sympathetic nerve: speed and reach **milliseconds, one target tissue, ended by reuptake**

Adrenal medulla: speed and reach **seconds to start, whole body, lasts minutes**

Why both? **the nerve gives fast local control; the hormone gives the whole-body fight or flight response and reaches tissues no nerve supplies**

Extra notes and drawings

BIO 005 Week 9 · The Heart as a Pump

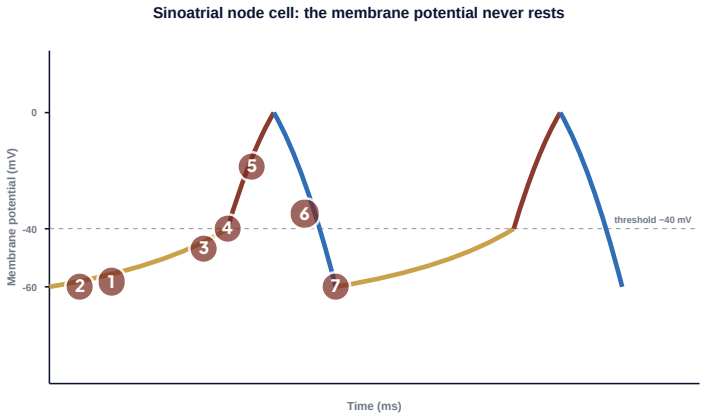
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Name _____

D. The pacemaker and the cardiac action potential

YOU SHOULD BE ABLE TO

Pacemaker potential. Explain the unstable pacemaker potential of the sinoatrial node by the funny current, calcium entry, and potassium exit and state what sets intrinsic heart rate.
Contractile cell action potential. Diagram the ventricular action potential and identify the ion movement responsible for the rapid upstroke, the plateau, and repolarization.
Cardiac refractory period. Explain why the long refractory period of cardiac muscle prevents tetanus and why that matters for pump function.



SA node cell: pacemaker potential

1. Pacemaker potential (slow depolarization)
2. Funny current, I_f
3. T-type Ca^{2+} channels (SA node)
4. Threshold in the SA node
5. Upstroke of the SA node action potential
6. Repolarization of the SA node cell
7. Minimum potential (about −60 mV)

WHAT YOU ARE LOOKING AT

Membrane potential of one SA node cell over two heartbeats. Notice there is no flat resting line: as soon as one beat ends the cell starts drifting toward the next.

Extra notes and drawings

BIO 005 Week 9 · The Heart as a Pump

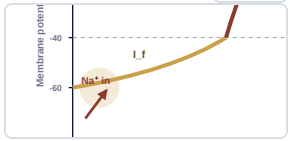
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D. The pacemaker and the cardiac action potential (continued)

HOW IT HAPPENS, IN ORDER (NUMBERS MATCH THE GOLD MARKERS ON THE SLIDE)

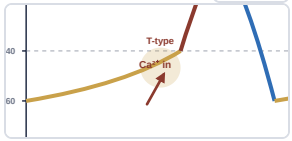
STEP 1



At −60 mV the funny channels open

HCN channels are unusual: they open when the membrane is hyperpolarized, not depolarized, and cAMP makes them open more. They pass mostly Na^+ in (a little K^+ out), so the cell drifts upward: the funny current, I_f

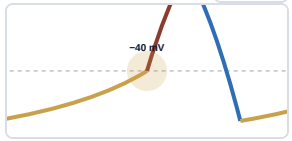
STEP 2



T-type Ca^{2+} channels add to the climb

As the cell nears threshold the small, transient (T-type) voltage-gated Ca^{2+} channels open briefly and Ca^{2+} entry finishes the pacemaker potential

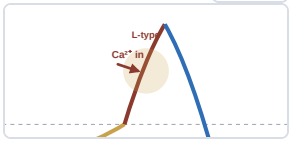
STEP 3



Threshold, about −40 mV

The long-lasting (L-type) voltage-gated Ca^{2+} channels open; these are the same channel type a calcium channel blocker acts on

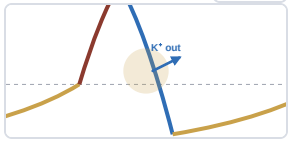
STEP 4



Upstroke: Ca^{2+} in

Depolarization to about 0 mV carried by Ca^{2+}, not Na^+; there are no fast Na^+ channels here

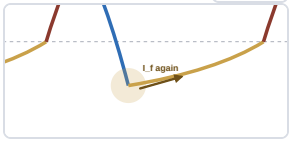
STEP 5



Voltage-gated K^+ channels open: repolarization

The L-type Ca^{2+} channels inactivate, delayed-rectifier K^+ channels open, K^+ leaves down its gradient and the potential falls back to −60 mV

STEP 6

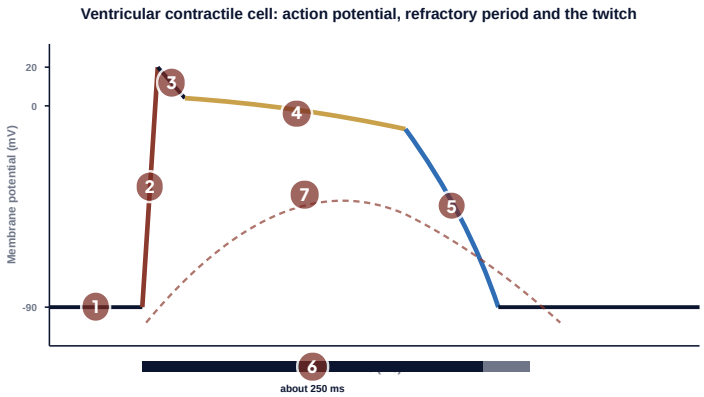


Back at −60 mV, it starts again

K^+ channels close, the funny channels reopen; this is the next beat starting

THE EFFECT: WHAT CHANGES BECAUSE OF THIS

The slope of the pacemaker potential sets heart rate. Sympathetic cAMP steepens it (faster); parasympathetic ACh flattens it and drops the floor (slower). That is the whole mechanism of heart rate control, and the SA node wins because its slope is steepest.



1. Phase 4: resting membrane potential (ventricle)
2. Phase 0: rapid upstroke
3. Phase 1: brief early repolarization
4. Phase 2: plateau
5. Phase 3: repolarization (ventricle)
6. Absolute refractory period
7. Tension curve (cardiac twitch)

WHAT YOU ARE LOOKING AT

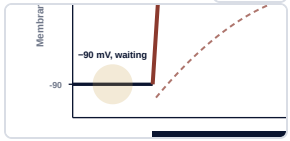
One ventricular contractile cell: membrane potential in solid color, the twitch tension dashed, and the refractory period as the bar underneath. Numbers on the left are millivolts.

Ventricular contractile cell: action potential and twitch

D. The pacemaker and the cardiac action potential (continued)

HOW IT HAPPENS, IN ORDER (NUMBERS MATCH THE GOLD MARKERS ON THE SLIDE)

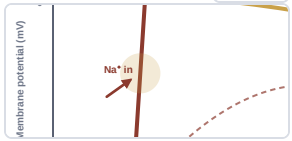
STEP 1



Phase 4: resting at −90 mV

K^+ leak holds it; this cell waits for a neighbor, it does not drift

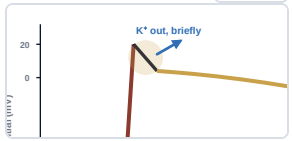
STEP 2



Phase 0: Na^+ rushes in

A neighbor depolarizes this cell to threshold through gap junctions; fast voltage-gated Na^+ channels open and Na^+ pours in; the upstroke takes 1 to 2 ms

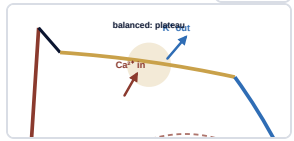
STEP 3



Phase 1: a brief dip

Na^+ channels inactivate and a few K^+ channels open

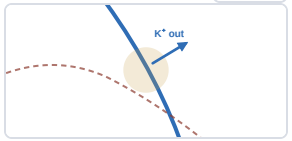
STEP 4



Phase 2: the plateau, Ca^{2+} in

Voltage-gated L-type Ca^{2+} channels (opened by the depolarization) let Ca^{2+} in while K^+ permeability is low; the two currents balance for about 200 ms. This Ca^{2+} is the trigger for calcium-induced calcium release and the contraction

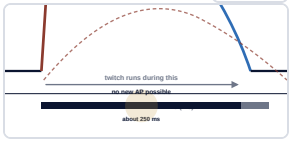
STEP 5



Phase 3: K^+ out, repolarization

The Ca^{2+} channels inactivate and the slow delayed-rectifier K^+ channels open; K^+ leaves and the cell returns to −90 mV. On the ECG this is the T wave

STEP 6



Refractory through most of the twitch

Na^+ channels stay inactivated about 250 ms, so no second action potential can start until the contraction is nearly over

THE EFFECT: WHAT CHANGES BECAUSE OF THIS

The plateau does two jobs: it brings in the Ca^{2+} that starts the contraction, and it keeps the cell refractory until the twitch is almost finished. So cardiac muscle cannot be tetanized, the ventricle always relaxes and refills, and the pump keeps pumping.

RATES AND THE REFRACTORY PERIOD

SA node intrinsic rate	**about 90 to 100 a minute on its own; about 70 with resting vagal tone**
AV node backup rate	**40 to 60 a minute**
Purkinje fiber backup rate	**25 to 40 a minute**
Why the heart cannot tetanize	**the refractory period (about 250 ms) lasts almost as long as the twitch, so a second contraction cannot stack on the first**
Why that matters	**the ventricle must relax to fill; a tetanized heart would stop pumping**

Extra notes and drawings

BIO 005 Week 9 · The Heart as a Pump

Smooth and cardiac muscle, the pacemaker, the wiring, the ECG, the cycle, and what sets cardiac output. Fill in as we go. Small square by a picture: scan it to see that picture full size on your phone.

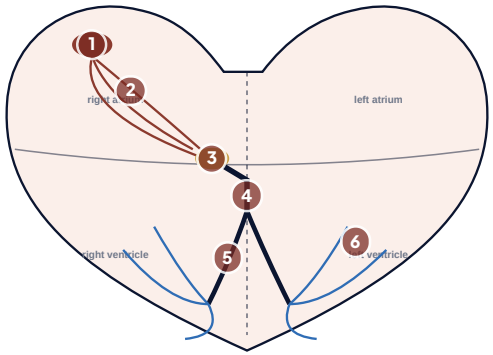
Name _____

E. The wiring and the ECG

YOU SHOULD BE ABLE TO

Conduction system. Trace an impulse through the sinoatrial node, internodal pathways, atrioventricular node, bundle of His, bundle branches, and Purkinje fibers and explain the purpose of the atrioventricular delay.
ECG waves and intervals. Label the P wave, QRS complex, T wave, PR interval, and QT interval and state the electrical event each represents.
ECG interpretation. Calculate heart rate from an ECG strip and identify sinus rhythm, tachycardia, bradycardia, and a first degree or complete heart block.
ECG recording. Record a lead II ECG, measure the intervals, and correlate each wave with the mechanical event that follows it.

The heart's wiring: where the signal starts and where it goes



Conduction system of the heart

1. Sinoatrial (SA) node
2. Internodal pathways
3. Atrioventricular (AV) node
4. Bundle of His (AV bundle)
5. Bundle branches
6. Purkinje fibers

WHAT YOU ARE LOOKING AT

The heart from the front, so the patient's right is on your left. Pacemaker and conducting tissue are in color; the dashed line is the septum and the faint curve divides atria from ventricles.

Extra notes and drawings

BIO 005 Week 9 · The Heart as a Pump

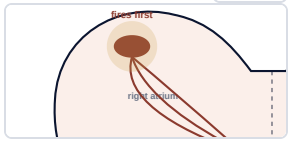
Smooth and cardiac muscle, the pacemaker, the wiring, the ECG, the cycle, and what sets cardiac output. Fill in as we go. Small square by a picture: scan it to see that picture full size on your phone.

Name _____

E. The wiring and the ECG (continued)

HOW IT HAPPENS, IN ORDER (NUMBERS MATCH THE GOLD MARKERS ON THE SLIDE)

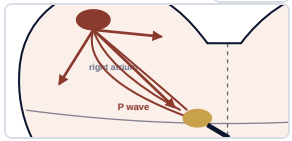
STEP 1



The SA node fires

The pacemaker with the steepest slope sets the pace, about 70 a minute

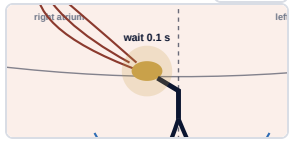
STEP 2



The atria depolarize

Along the internodal pathways and cell to cell through gap junctions: the P wave; atrial contraction follows

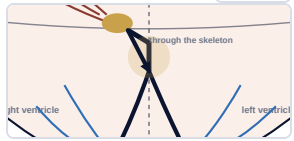
STEP 3



The AV node holds the signal about 0.1 s

The only road through the fibrous skeleton is slow on purpose, so the atria finish emptying first

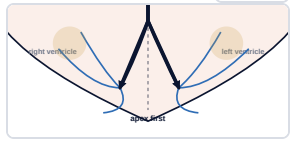
STEP 4



Bundle of His, then the bundle branches

Fast conduction down the septum toward the apex

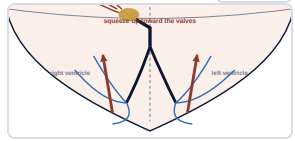
STEP 5



Purkinje fibers reach every ventricular cell

Within about 0.1 s the whole ventricle depolarizes: the QRS

STEP 6



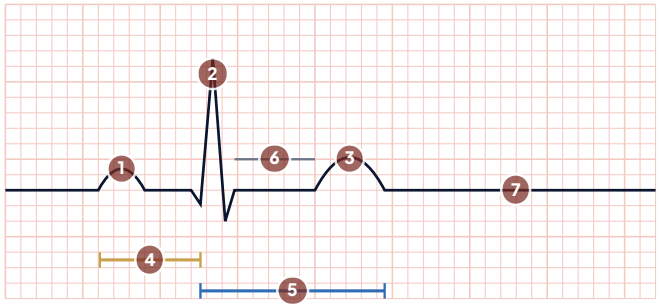
The ventricles squeeze from the apex up

Blood is pushed toward the valves at the top; then the ventricles repolarize: the T wave

THE EFFECT: WHAT CHANGES BECAUSE OF THIS

Atria first, a pause, then the ventricles from the bottom up: that order is what makes the pump efficient. Every wave on the ECG is one of these steps seen from the skin, and a block at any point shows up as a stretched or missing connection between P and QRS.

One heartbeat on paper: lead II



Small box 0.04 s, big box 0.20 s · Heart rate = 1500 ÷ small boxes between R waves, or 300 ÷ big boxes

Lead II, one beat

1. P wave
2. QRS complex
3. T wave
4. PR interval
5. QT interval
6. ST segment
7. Isoelectric baseline

WHAT YOU ARE LOOKING AT

Lead II, one beat, on standard ECG paper. A small box is 0.04 s, a big box 0.20 s. The flat line is the baseline; every wave is read against it.

BIO 005 Week 9 · The Heart as a Pump

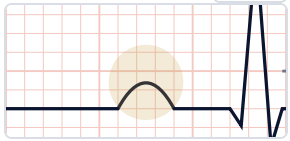
Smooth and cardiac muscle, the pacemaker, the wiring, the ECG, the cycle, and what sets cardiac output. Fill in as we go. Small square by a picture: scan it to see that picture full size on your phone.

Name _____

E. The wiring and the ECG (continued)

HOW IT HAPPENS, IN ORDER (NUMBERS MATCH THE GOLD MARKERS ON THE SLIDE)

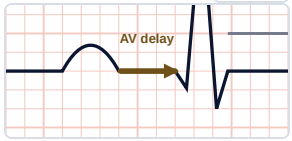
STEP 1



P wave: the atria depolarize

The SA node signal spreading across both atria; atrial contraction follows

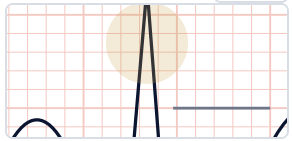
STEP 2



PR segment: the AV node delay

Flat line while the signal waits in the AV node; PR interval normally 0.12 to 0.20 s

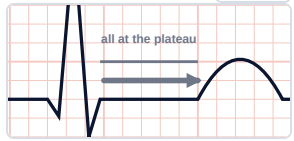
STEP 3



QRS: the ventricles depolarize

Fast, under 0.12 s, because of the Purkinje system; ventricular contraction follows

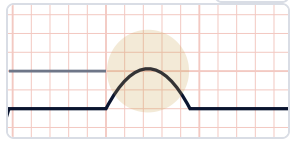
STEP 4



ST segment: all depolarized

The plateau of every ventricular cell at once; a shifted ST means injured muscle

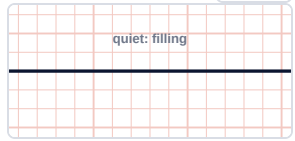
STEP 5



T wave: the ventricles repolarize

Phase 3 of every cell; ventricular relaxation follows

STEP 6



Baseline: rest and filling

Nothing moving electrically; the ventricles are filling for the next beat

THE EFFECT: WHAT CHANGES BECAUSE OF THIS

The ECG is the electrical story of one beat, and each mechanical event follows its wave by a fraction of a second. Rate comes from the spacing of the R waves; a stretched PR is first-degree block; P waves marching on their own with slow wide QRS complexes is complete block.

READ THE STRIP

Heart rate from the strip	**1500 ÷ small boxes between two R waves, or 300 ÷ big boxes; or count beats in 6 seconds × 10**
Tachycardia	**sinus rhythm faster than 100 a minute**
First-degree heart block	**every P is followed by a QRS, but the PR interval is longer than 0.20 s**
Wave to mechanical event	**P then atrial contraction; QRS then ventricular contraction; T then ventricular relaxation**
Sinus rhythm	**a P wave before every QRS, a QRS after every P, regular, 60 to 100 a minute**
Bradycardia	**sinus rhythm slower than 60 a minute**
Complete (third-degree) heart block	**P waves and QRS complexes march independently; the ventricles run on a Purkinje escape rate of 25 to 40**

Extra notes and drawings

BIO 005 Week 9 · The Heart as a Pump

Smooth and cardiac muscle, the pacemaker, the wiring, the ECG, the cycle, and what sets cardiac output. Fill in as we go. Small square by a picture: scan it to see that picture full size on your phone.

Name _____

F. The cardiac cycle, the loop, and the sounds

YOU SHOULD BE ABLE TO

Cardiac cycle. Order the phases of the cardiac cycle and state the pressure relationship that opens and closes the atrioventricular and semilunar valves in each phase.
Pressure volume loop. Label the four phases of a ventricular pressure volume loop and predict how the loop changes with increased preload, increased afterload, or increased contractility.
Heart sounds. Explain the origin of the first and second heart sounds and relate a murmur to valve stenosis or regurgitation by its timing.
Heart sound and pulse correlation. Auscultate the heart sounds, locate the valve areas, and correlate the sounds with the pulse and with the cardiac cycle phases.

THE CARDIAC CYCLE, PHASE BY PHASE

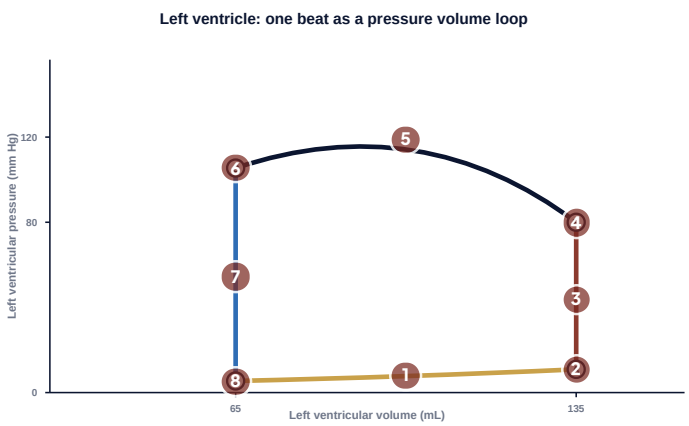
STEP 1
Atrial systole
Atria contract and top up the ventricles to EDV; AV valves open, semilunar valves closed

STEP 2
Isovolumic ventricular contraction
Ventricular pressure rises above atrial pressure: AV valves close, S1. Not yet above aortic pressure, so all valves closed and volume fixed

STEP 3
Ventricular ejection
Ventricular pressure passes aortic and pulmonary pressure: semilunar valves open, blood leaves, volume falls to ESV

STEP 4
Isovolumic ventricular relaxation
Pressure falls below aortic pressure: semilunar valves close, S2. Still above atrial pressure, so all valves closed

STEP 5
Ventricular filling
Pressure falls below atrial pressure: AV valves open, blood pours in passively; most filling happens here



Left ventricular pressure volume loop

1. Ventricular filling (PV loop)
2. Point A: end-diastolic volume, mitral valve closes
3. Isovolumic contraction
4. Point B: aortic valve opens
5. Ventricular ejection
6. Point C: end-systolic volume, aortic valve closes
7. Isovolumic relaxation
8. Point D: mitral valve opens

WHAT YOU ARE LOOKING AT

Left ventricular pressure (up) against volume (across) for one beat. Time runs counterclockwise around the loop; the corners are the four valve events.

Extra notes and drawings

BIO 005 Week 9 · The Heart as a Pump

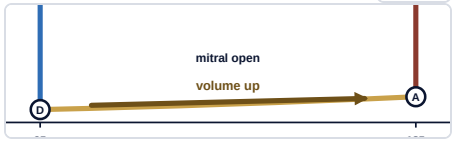
Smooth and cardiac muscle, the pacemaker, the wiring, the ECG, the cycle, and what sets cardiac output. Fill in as we go. Small square by a picture: scan it to see that picture full size on your phone.

Name _____

F. The cardiac cycle, the loop, and the sounds (continued)

HOW IT HAPPENS, IN ORDER (NUMBERS MATCH THE GOLD MARKERS ON THE SLIDE)

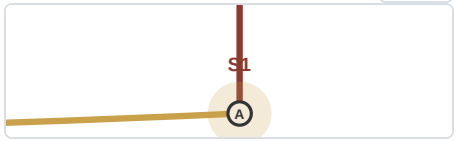
STEP 1



Filling: mitral open, pressure low

Volume climbs from ESV to EDV, most of it passive, the last bit from atrial contraction

STEP 2



A: the mitral valve closes

Ventricular pressure rises above atrial pressure: S1. Volume here is EDV, the preload

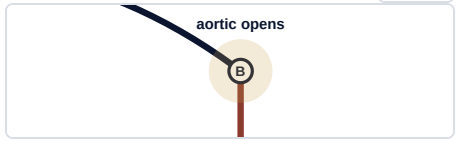
STEP 3



Isovolumic contraction

All valves closed, pressure climbs, volume cannot change

STEP 4



B: the aortic valve opens

Ventricular pressure passes aortic pressure, about 80 mm Hg: the afterload

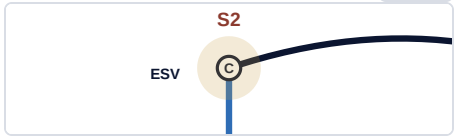
STEP 5



Ejection

Blood leaves, volume falls from EDV to ESV, pressure peaks near 120 mm Hg

STEP 6



C: the aortic valve closes

Pressure drops below aortic pressure: S2. Volume here is ESV

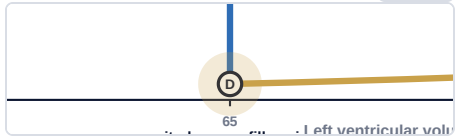
STEP 7



Isovolumic relaxation

All valves closed again while pressure falls

STEP 8



D: the mitral valve opens

Pressure drops below atrial pressure and filling starts over

THE EFFECT: WHAT CHANGES BECAUSE OF THIS

The width of the loop is the stroke volume. More preload moves A to the right and widens it (Frank–Starling). More afterload lifts B and narrows it. More contractility pulls C to the left and widens it. A failing ventricle makes a short, narrow loop.

Extra notes and drawings

BIO 005 Week 9 · The Heart as a Pump

Smooth and cardiac muscle, the pacemaker, the wiring, the ECG, the cycle, and what sets cardiac output. Fill in as we go. Small square by a picture: scan it to see that picture full size on your phone.

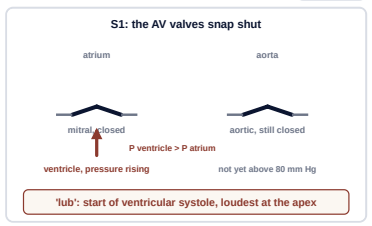
Name _____

F. The cardiac cycle, the loop, and the sounds (continued)

WHAT YOU HEAR, AND WHEN

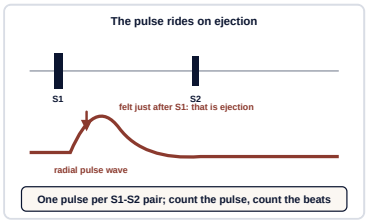
Heart sounds

S1, 'lub'



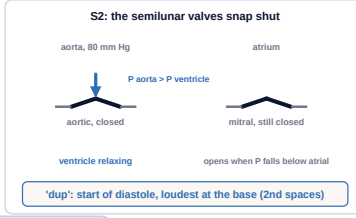
AV valves (mitral and tricuspid) closing at the start of ventricular systole

The pulse



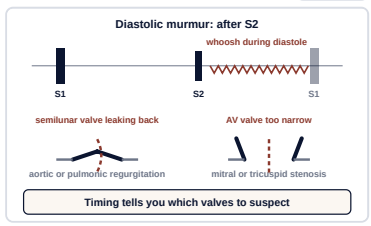
Felt just after S1, during ejection; one pulse per S1–S2 pair

S2, 'dup'



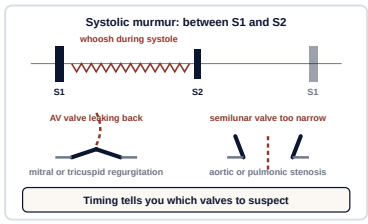
…lves (aortic and pulmonary) …e start of diastole

…urmur



After S2: a semilunar valve leaking or an AV valve too narrow

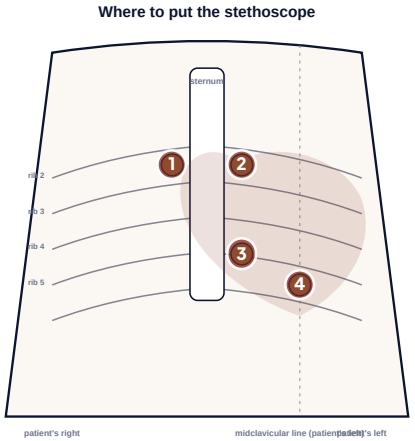
Systolic murmur



Between S1 and S2: an AV valve leaking (regurgitation) or a semilunar valve too narrow (stenosis)

Extra notes and drawings

F. The cardiac cycle, the loop, and the sounds (continued)



Valve areas on the chest wall

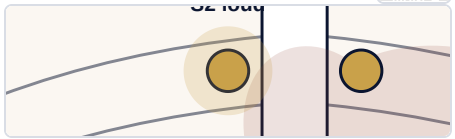
1. **Aortic area**
2. **Pulmonic area**
3. **Tricuspid area**
4. **Mitral area (apex)**

WHAT YOU ARE LOOKING AT

The front of the chest: sternum in the middle, ribs counted from the top, the heart faint behind. Sound travels in the direction the blood flows, so each valve is heard a little downstream of where it sits.

HOW IT HAPPENS, IN ORDER (NUMBERS MATCH THE GOLD MARKERS ON THE SLIDE)

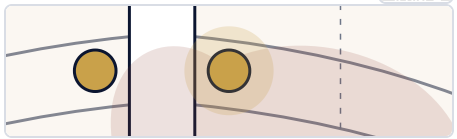
STEP 1



Aortic area: right 2nd space

S2 is loud here; aortic stenosis gives a harsh systolic murmur

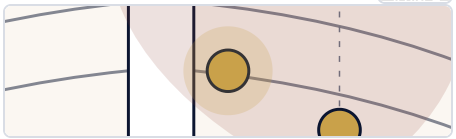
STEP 2



Pulmonic area: left 2nd space

The other S2 valve

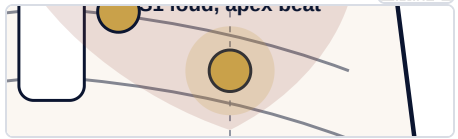
STEP 3



Tricuspid area: left lower sternal border

S1 from the right side; tricuspid regurgitation is a systolic murmur here

STEP 4



Mitral area: 5th space, midclavicular

The apex; S1 loudest, and the apical pulse is felt here

THE EFFECT: WHAT CHANGES BECAUSE OF THIS

Listen at each site for the two sounds and anything between them. A murmur between S1 and S2 is systolic (an AV valve leaking or a semilunar valve too narrow); after S2 it is diastolic (the reverse). Feel the radial pulse while you listen: it arrives just after S1.

BIO 005 Week 9 · The Heart as a Pump

Smooth and cardiac muscle, the pacemaker, the wiring, the ECG, the cycle, and what sets cardiac output. Fill in as we go. Small square by a picture: scan it to see that picture full size on your phone.

Name _____

G. Stroke volume, cardiac output, and what controls them

YOU SHOULD BE ABLE TO

Stroke volume and ejection fraction. Calculate stroke volume and ejection fraction from end diastolic and end systolic volumes and interpret a reduced ejection fraction.
Cardiac output. Calculate cardiac output and cardiac reserve and predict the effect of a change in heart rate or stroke volume on each.
Frank Starling relationship. State the Frank Starling law and explain how venous return and end diastolic volume determine the force of the next contraction.
Preload afterload and contractility. Distinguish preload, afterload, and contractility and predict how a change in each alters stroke volume.
Autonomic regulation of the heart. Explain how sympathetic and parasympathetic input alter heart rate and contractility at the receptor and second messenger level.

THE NUMBERS (TYPICAL RESTING ADULT)

End-diastolic volume (EDV) **about 135 mL**

End-systolic volume (ESV) **about 65 mL**

Stroke volume = EDV − ESV **135 − 65 = 70 mL per beat**

Ejection fraction = SV ÷ EDV **70 ÷ 135 = 52%; under about 40% means the ventricle is failing to empty (reduced ejection fraction)**

Cardiac output = HR × SV **72 × 70 mL = 5.0 L/min, about the whole blood volume every minute**

Cardiac reserve **maximum CO minus resting CO; a trained heart can reach 20 to 25 L/min**

Change HR or SV **CO rises or falls in proportion; raising both at once is how exercise gets a fourfold increase**

Extra notes and drawings

BIO 005 Week 9 · The Heart as a Pump

Smooth and cardiac muscle, the pacemaker, the wiring, the ECG, the cycle, and what sets cardiac output. Fill in as we go. Small square by a picture: scan it to see that picture full size on your phone.

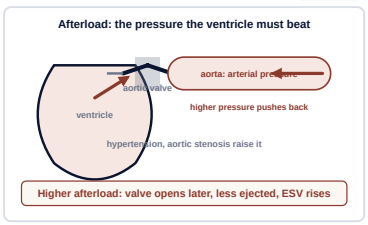
Name _____

G. Stroke volume, cardiac output, and what controls them (cont

WHAT SETS CARDIAC OUTPUT?

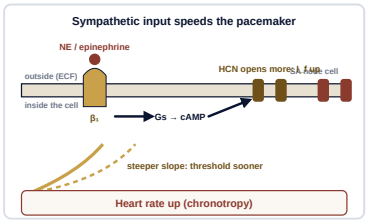
Cardiac output = HR × SV

Afterload



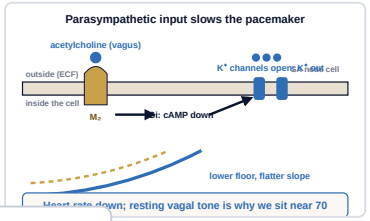
The arterial pressure the ventricle must beat to open the semilunar valve; higher afterload means lower SV at the same force

Heart rate: sympathetic up



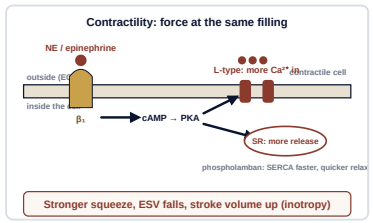
Norepinephrine and epinephrine on β_1 receptors in the SA node speed the pacemaker potential

Heart rate: parasympathetic down



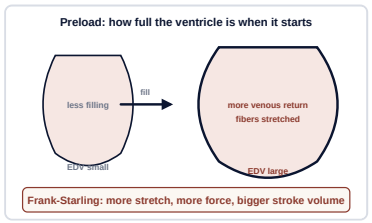
e on M_2 receptors opens K^+ l slows the drift to threshold

ty



Force at a given EDV; raised by sympathetic input and epinephrine through β_1 and more Ca^{2+}

Preload



End-diastolic volume, set by venous return; more stretch means more force (Frank-Starling)

SYMPATHETIC INPUT AT THE RECEPTOR LEVEL

STEP 1
Norepinephrine or epinephrine binds β_1
A G protein-coupled receptor on pacemaker and contractile cells

STEP 2
G_s activates adenylyl cyclase
Cytoplasmic cAMP rises

STEP 3
In the SA node, cAMP acts on HCN channels
More I_f, steeper pacemaker potential, faster rate (chronotropy)

STEP 4
In contractile cells, protein kinase A phosphorylates targets
L-type Ca^{2+} channels let in more Ca^{2+}, more SR release, stronger contraction (inotropy)

STEP 5
PKA also phosphorylates phospholamban
SERCA pumps Ca^{2+} back faster, so relaxation is quicker and there is time to fill at high rates

STEP 6
Parasympathetic does the reverse at the SA node
ACh on M_2 receptors: G_i opens K^+ channels (hyperpolarizes) and lowers cAMP; little direct effect on ventricular force

G. Stroke volume, cardiac output, and what controls them (continued)

FRANK-STARLING IN ONE LINE EACH

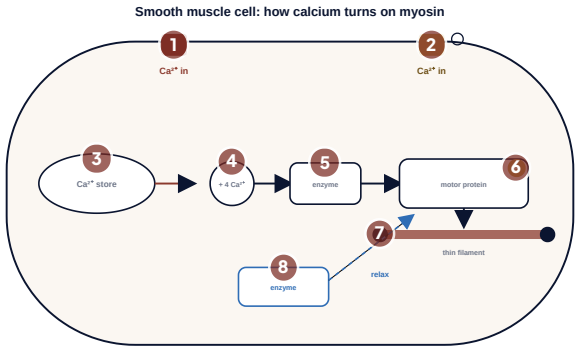
The law	**the more the ventricle is stretched by filling (EDV), the harder it contracts, so stroke volume matches venous return**
Why	**at longer sarcomere length more crossbridges can form and the filaments are more sensitive to Ca^{2+}**
What raises venous return	**skeletal muscle pump, respiratory pump, sympathetic venoconstriction, more blood volume**
Why it matters	**the two sides of the heart stay matched beat to beat without any nerve telling them to**

Extra notes and drawings

BIO 005 Week 9 · The Heart as a Pump

A. Smooth muscle: how it contracts and what controls it · Competency Smooth muscle contraction mechanism · Smooth muscle types and regulation · one card per numbered part, the same three lines the slide reveals. Write the name, then what it is, what it does and why it matters, as we go.

Name _____



Smooth muscle cell: the calcium switch

1 Voltage-gated Ca^{2+} channel

What it is The electrical door: an L-type channel that opens when the membrane depolarizes

What triggers it Anything that moves the voltage: stretch opening stretch-sensitive cation channels, a pacemaker slow wave arriving through gap junctions, high extracellular K^+, or a transmitter that opens a cation channel or closes a K^+ channel

What it does Lets Ca^{2+} in from the extracellular fluid, the main Ca^{2+} source for smooth muscle

Why it matters Calcium channel blockers shut only this door, so they relax vessels and lower blood pressure without stopping tissues that run on the receptor route

2 Receptor-operated Ca^{2+} channel

What it is The chemical door: Ca^{2+} entry and SR release started by a ligand on a G protein-coupled receptor, not by voltage

What triggers it Norepinephrine on α_1, acetylcholine on M_3, angiotensin II, oxytocin, endothelin: the receptor activates phospholipase C, IP_3 opens the SR Ca^{2+} channels, and receptor-operated channels in the membrane open

What it does Lets Ca^{2+} rise with no action potential at all (pharmacomechanical coupling)

Why it matters Which door a tissue uses is set by the receptors and channels it was built with: stretch and pacemakers use the voltage door, hormones the receptor door, nerves usually both, since norepinephrine also closes K^+ channels and depolarizes the cell

3 Sarcoplasmic reticulum (smooth muscle)

What it is A small internal Ca^{2+} store, much less developed than in skeletal muscle

What it does Releases Ca^{2+} through IP_3 receptors and ryanodine receptors when signaled

Why it matters Smooth muscle leans on extracellular Ca^{2+} because this store is small

6 Phosphorylated myosin light chain

What it is Myosin with a phosphate on its regulatory light chain

What it does Only phosphorylated myosin can bind actin and cycle

Why it matters Dephosphorylated myosin can stay attached in the latch state: tension held with almost no ATP

7 Dense bodies

What it is Protein anchors in the cytoplasm and on the membrane, the smooth muscle version of Z discs

What it does Actin filaments attach here, so shortening pulls the whole cell in like a drawstring

Why it matters No sarcomeres, no striations, and the cell can shorten far more than skeletal muscle

8 Myosin light chain phosphatase

What it is The enzyme that removes the phosphate from the myosin light chain

What it does Ends crossbridge cycling as Ca^{2+} falls and calmodulin releases MLCK

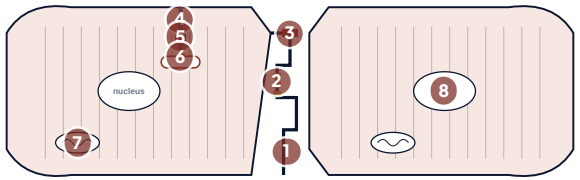
Why it matters Signals that change this enzyme's activity change tone without changing Ca^{2+} (Ca^{2+} sensitization)

BIO 005 Week 9 · The Heart as a Pump

B. Cardiac muscle. Find the three muscle types side by side · Competency Cardiac muscle properties · Comparison of the three muscle types · one card per numbered part the same lines the slide reveals. Write the name, then what it is, what it does and why it matters, as we go.

Name _____

Two cardiac muscle cells joined at an intercalated disc



Two cardiac cells and the intercalated disc

1 Intercalated disc

What it is The stepped junction where one cardiac cell ends and the next begins

What it does Holds the cells together and carries the electrical signal across

Why it matters Because of it the myocardium works as one unit, a functional syncytium

2 Gap junction

What it is Channels (connexons) in the intercalated disc that join the cytoplasm of two cells

What it does Let ions and the action potential pass straight to the next cell

Why it matters This is why one SA node signal spreads to every atrial cell; the ventricles form a second syncytium

3 Desmosome

What it is A strong protein anchor in the intercalated disc

What it does Holds cells together through the pull of every beat

Why it matters Without it the cells would tear apart under cardiac force

4 Calmodulin

What it is The Ca^{2+}-binding protein of smooth muscle; it takes the place troponin has in striated muscle

What it does Binds four Ca^{2+}, then activates myosin light chain kinase

Why it matters Regulation is on the thick filament (myosin), not on actin

6 Ryanodine receptor (cardiac SR)

What it is The Ca^{2+} release channel on the sarcoplasmic reticulum

What it does Opens when trigger Ca^{2+} binds it, releasing the SR store, about 90% of the Ca^{2+} for contraction

Why it matters Cardiac EC coupling needs both ECF and SR Ca^{2+}; skeletal muscle needs SR only

7 Mitochondrion (cardiac)

What it is Up to a third of the cardiac cell's volume is mitochondria

What it does Makes ATP aerobically, all day, from fatty acids, glucose and lactate

Why it matters The heart cannot run an oxygen debt; cut its oxygen and the muscle dies (infarction)

8 Branching, single nucleus

What it is Cardiac cells are short, branched and usually have one central nucleus

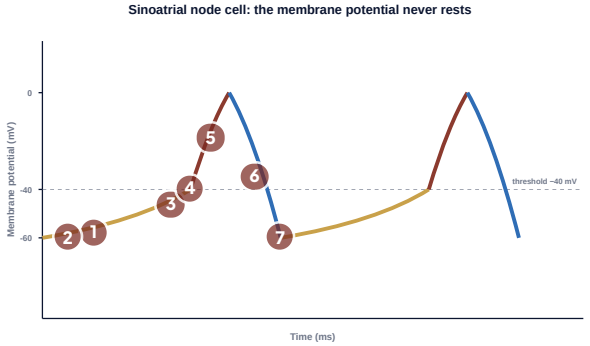
What it does Branches let each cell touch several neighbors

Why it matters Tells cardiac apart from skeletal (long, many nuclei at the edge) and smooth (spindle, one nucleus, no striations)

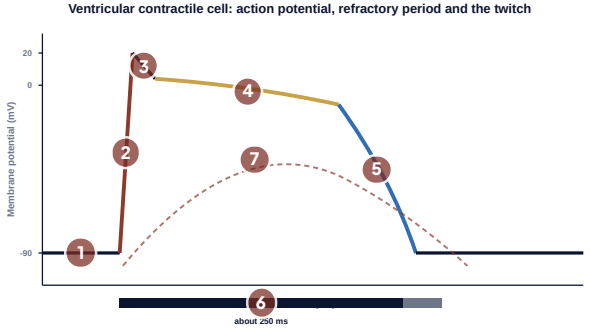
BIO 005 Week 9 · The Heart as a Pump

D. The pacemaker and the cardiac action potential · Competency Pacemaker potential · Contractile cell action potential · Cardiac refractory period, one card per numbered part, the same three lines the slide reveals. Write the name, then what it is, what it does and why it matters as we go.

Name _____



SA node cell: pacemaker potential



Ventricular contractile cell: action potential and twitch

1 Pacemaker potential (slow depolarization)

What it is The slow upward drift from about −60 mV to threshold between beats

What it does Makes the SA node fire on its own, with no nerve input

Why it matters Its slope sets heart rate: steeper slope, faster heart

2 T-type Ca²⁺ channels

What it is An inward fold of the cell membrane, larger than in skeletal muscle

What it does Carries the action potential deep into the cell to the L-type Ca^{2+} channels

Why it matters This is where the electrical signal first lets Ca^{2+} in

3 T-type Ca²⁺ channels (SA node)

What it is Transient Ca^{2+} channels that open as the cell nears threshold

What it does Add Ca^{2+} entry that finishes the climb to threshold

Why it matters Pacemaker depolarization is driven by Na^+ and Ca^{2+}, not by Na^+ alone

4 Threshold in the SA node

What it is About −40 mV in a pacemaker cell

What it does Reaching it opens the L-type Ca^{2+} channels

Why it matters The SA node reaches threshold first (about 70 times a minute with vagal tone), so it sets the rhythm

5 Upstroke of the SA node action potential

What it is The rapid rise to about 0 mV

What it does Carried by Ca^{2+} entering through L-type channels, not Na^+

Why it matters Pacemaker cells have no fast Na^+ channels, so the upstroke is slower than in a ventricular cell

6 Repolarization of the SA node cell

What it is The fall back toward −60 mV

What it does K^+ leaves through voltage-gated K^+ channels as the Ca^{2+} channels close

Why it matters As soon as the K^+ channels close, I_f starts the next pacemaker potential

7 Minimum potential (about −60 mV)

What it is The lowest point of the cycle; there is no true resting potential

What it does Hyperpolarization here is what opens the HCN channels

Why it matters Parasympathetic K^+ channels push this lower, so the climb to threshold takes longer: slower heart

4 Phase 2: plateau

What it is The long flat part, about 200 ms, near 0 mV

What it does Ca^{2+} in through L-type channels balances K^+ out; this Ca^{2+} triggers contraction

Why it matters The plateau makes the cardiac action potential as long as the twitch, which prevents tetanus

5 Phase 3: repolarization (ventricle)

What it is The return to −90 mV

What it does Ca^{2+} channels close, slow K^+ channels open, K^+ leaves

Why it matters On the ECG this is the T wave

6 Absolute refractory period

What it is About 250 ms during which no second action potential can start, because the Na^+ channels are inactivated

What it does Lasts almost as long as the contraction itself

Why it matters No summation, no tetanus: the ventricle relaxes and fills between beats, which is the whole point of a pump

7 Tension curve (cardiac twitch)

What it is The contraction that follows the action potential

What it does Rises and falls while the cell is still refractory

Why it matters In skeletal muscle the refractory period is over before tension peaks, so skeletal muscle can be tetanized and cardiac cannot

BIO 005 Week 9 · The Heart as a Pump

E. The wiring and the ECG · competency: conduction system · ECG waves and intervals · ECG interpretation · ECG recording · one card per numbered part, the same three lines the slide reveals. Write the name, then what it is, what it does and why it matters, as we go.

Name _____

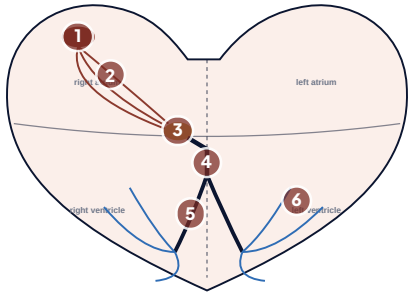
4 Funny channels

What it is A current through HCN channels that open when the cell hyperpolarizes (the opposite of most channels)

What it does Lets Na^+ in, starting the pacemaker potential

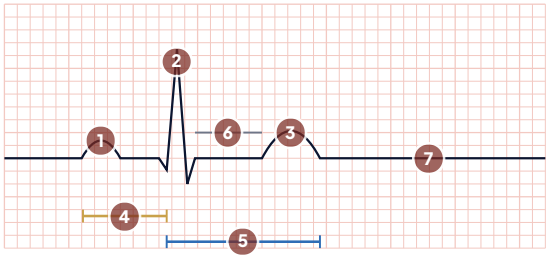
Why it matters Sympathetic cAMP opens more of these channels; the drug

The heart's wiring: where the signal starts and where it goes



Conduction system of the heart

One heartbeat on paper: lead II



Small box 0.04 s, big box 0.20 s · Heart rate = 1500 ÷ small boxes between R waves, or 300 ÷ big boxes

Lead II, one beat

1 Sinoatrial (SA) node

What it is A patch of pacemaker cells in the right atrial wall near the superior vena cava

What it does Fires first, about 70 times a minute, and starts every normal beat

Why it matters A rhythm set by the SA node is called sinus rhythm

2 Internodal pathways

What it is Conducting fibers running through the atria from SA node to AV node

What it does Spread the signal across both atria (P wave) and deliver it to the AV node

Why it matters The fibrous skeleton stops the signal there: the AV node is the only electrical road to the ventricles

3 Pha[redacted]mbrane potential (ventricle)

What it is A stable −90 mV between beats

What it does Held by K^+ leak; contractile cells wait for a signal from a neighbor

Why it matters Unlike the SA node, contractile cells do not drift to

3 Atrioventricular (AV) node

What it is Pacemaker and conducting tissue at the floor of the right atrium

What it does Slows the signal by about 0.1 s so the atria finish contracting before the ventricles start

Why it matters The AV delay is the PR segment; its backup rate is 40 to 60 a minute

4 Bundle of His (AV bundle)

What it is The conducting bundle that pierces the fibrous skeleton into the septum

What it does Carries the signal from the AV node to the bundle branches

Why it matters The one route through the insulating fibrous skeleton

5 Bundle branches

What it is Left and right branches running down the interventricular septum

What it does Carry the signal fast to the apex

Why it matters Depolarization starts at the apex, so the ventricles squeeze from the bottom up toward the outflow valves

6 Purkinje fibers

What it is Large, fast-conducting fibers that spread from the apex through the ventricular walls

What it does Deliver the signal to the contractile cells almost at once (QRS lasts under 0.12 s)

Why it matters Backup rate of 25 to 40 a minute if everything above fails

1 P wave

What it is The first small bump

What it does Atrial depolarization, spreading from the SA node

Why it matters Atrial contraction follows it; a P wave before every QRS is one mark of sinus rhythm

5 QT interval

What it is Start of QRS to end of T

What it does Covers ventricular depolarization and repolarization: one ventricular action potential

Why it matters A long QT raises the risk of dangerous ventricular arrhythmia

6 ST segment

What it is The flat stretch between the S wave and the T wave

What it does The ventricles are all depolarized (the plateau)

Why it matters Lifted or dropped above or below baseline points to injured heart muscle

7 Isoelectric baseline

What it is The flat line between beats

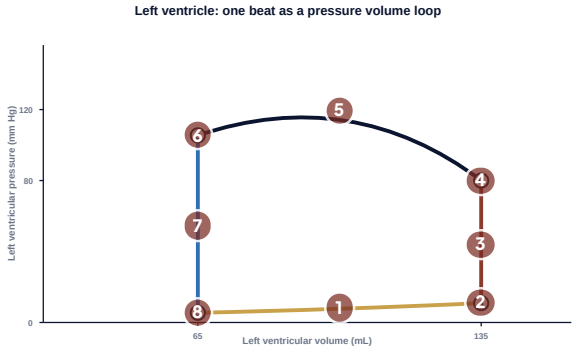
What it does No net electrical movement: the ventricles are at rest

Why it matters Every wave is measured against it

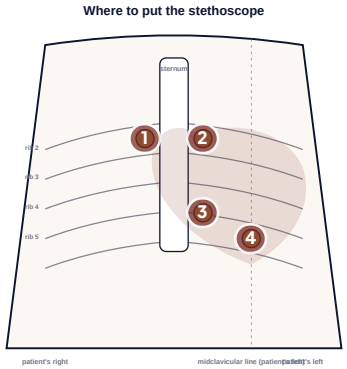
BIO 005 Week 9 · The Heart as a Pump

F. The cardiac cycle, the loop, and the sounds · Competency Cardiac cycle · Pressure volume loop · Heart sounds · Heart sound and pulse correlation · one card per numbered part, the same three lines the slide reveals. Write the name, then what it is, what it does and why it matters, as we go.

Name _____



Left ventricular pressure volume loop



Valve areas on the chest wall

1 Ventricular filling (PV loop)

What it is The bottom of the loop, running left to right

What it does Volume rises from ESV to EDV while pressure stays low; mitral valve open

Why it matters Most filling is passive; atrial contraction adds the last part

2 [redacted]

What it is The tall spike

What it does Ventricular depolarization (atrial repolarization hides inside it).

Why it matters Ventricular contraction follows; wider than 0.12 s means a slow route through the ventricles

3 Isovolumic contraction

What it is The straight line up on the right

What it does Pressure rises with all valves closed, so volume cannot change

Why it matters Pressure must beat aortic pressure (about 80 mm Hg) before anything leaves

4 Point B: aortic valve opens

What it is The top of the right side, about 80 mm Hg

What it does Ventricular pressure passes aortic pressure and ejection begins

Why it matters The afterload is the pressure the ventricle has to reach here; raise it and the loop gets taller and narrower

5 Ventricular ejection

What it is The top of the loop, running right to left

What it does Volume falls from EDV to ESV as blood leaves; pressure peaks near 120 mm Hg

Why it matters The horizontal width of this part is the stroke volume

6 Point C: end-systolic volume, aortic valve closes

What it is The top left corner, about 65 mL

What it does Ventricular pressure drops below aortic pressure; the aortic valve closes: S2

Why it matters ESV is what is left; stronger contractility pushes this corner left

7 Isovolumic relaxation

What it is The straight line down on the left

What it does Pressure falls with all valves closed

Why it matters Ends when pressure drops below atrial pressure

3 Tricuspid area

What it is Fourth or fifth intercostal space, left lower sternal border

What it does Best spot for the tricuspid valve

Why it matters Tricuspid regurgitation gives a systolic murmur here

4 Mitral area (apex)

What it is Fifth intercostal space, left midclavicular line: the apex

What it does S1 is loudest here; this is also where you feel the apical pulse

Why it matters Mitral stenosis gives a diastolic murmur here; mitral regurgitation a systolic one

2 Point A: end-diastolic volume, mitral valve closes

What it is The right-hand bottom corner, about 135 mL

What it does Ventricular pressure rises above atrial pressure, the mitral valve closes: S1

Why it matters EDV is the preload; raise it and the loop widens

8 Point D: mitral valve opens

What it is The bottom left corner

What it does Ventricular pressure falls below atrial pressure, the mitral valve opens, filling starts again

Why it matters The loop runs counterclockwise, one lap per beat

1 Aortic area

What it is Second intercostal space, right sternal border

What it does Best spot to hear the aortic valve, S2 loud here

Why it matters Aortic stenosis gives a harsh systolic murmur here

2 Pulmonic area

What it is Second intercostal space, left sternal border

What it does Best spot for the pulmonary valve

Why it matters Sound travels in the direction of flow, which is why the valve areas sit above the valves themselves